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Ring based latency-aware and energy-efficient Hybrid WDM TDM-PON with ODN interconnection capability for smart cities

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ABSTRACT

In the current scenario, there is a huge advancement in Information and Communication Technology (ICT), and almost most of the devices used in a smart city are the Internet of Things (IoT) based smart devices. To support high-speed IoT devices, wavelength division multiplexed passive optical networks (WDM-PON) is one of the most promising and most commonly used technology in optical distribution network (ODN). With the exponential growth of WDM-PON and its applications, energy consumption, operational expenditure (OPEX), communication latency between different users, reliability of service etc. are becoming the most challenging issues to implement the WDM-PON networks in the smart cities. In this paper, we propose a ring based latency aware and energy efficient PON architecture which is capable of providing point to point downstream, upstream, inter-ODN, intra-ODN and inter-ODN broadcasting transmission. Proposed architecture reduces the significant amount of transmission latency, energy consumption, operational expenditure and reduces the emission of Green House Gases (GHGs). The architecture also reduces the downstream and upstream bandwidth consumption by avoiding the transmission of the inter-ODN signal to optical line terminal (OLT) and back to destined ODN. The architecture seems appropriate for supporting various real-time high-speed ICT and IoT applications of a smart city.

1. Introduction

A smart city is a place that makes human life easier, smarter, and productive with the use of innovative technologies without imparting any negative consequences on human life. Nowadays, in smart cities, the lots of internet of things (IoT) and information and communication technology (ICT) applications are increasing exponentially [1]. Most of the applications are real-time technology-based and require high-speed internet connectivity. Few of the applications are in smart home atomization such as smart car parking, smart lights, smart cameras, robotic assistance for various applications etc. All these applications must be supported with very high bandwidth, low energy consumption, low latency, and higher service reliability. Energy consumption is increasing exponentially and there may be a situation in the near future where such energy demand may not be fulfilled. Furthermore, the larger energy consumption is responsible for emitting a significant amount of environmental unfriendly gasses like carbon monoxide (CO) or carbon dioxide (CO_2) etc. Therefore, high-bandwidth [2] and energy management [3] have become the two major challenging barriers to the development of smart cities.

Today, optical fiber technology is being accepted as one of the most suitable technology to transmit the data at higher speeds with lower energy consumption. Researchers have demonstrated up to 100 Gbit/s data rate transmission by developing the wavelength division multiplexed (WDM) passive optical networks (PONs). PONs have an optical line terminal (OLT) at the central office (CO) and multiple optical distribution networks (ODNs) at the user end. Further, a particular ODN provides the services to the optical networking units (ONUs)/end users. In the PON technology, a particular signal can be sent from OLT to the ODN as a downstream signal and from ODN to OLT as an upstream signal. Also, for various local applications, there is a need for frequent communication among various ODNs as one user may communicate many times with other users. For communication between two users, conventionally, the transmitting user initially sends the data to the OLT, and then this data is re-processed, re-modulated, and re-transmitted to the destined user. In this process, the signal needs to be first transmitted as an upstream signal and then as a downstream signal. So the two-time transmission of the same data is liable for more power consumption

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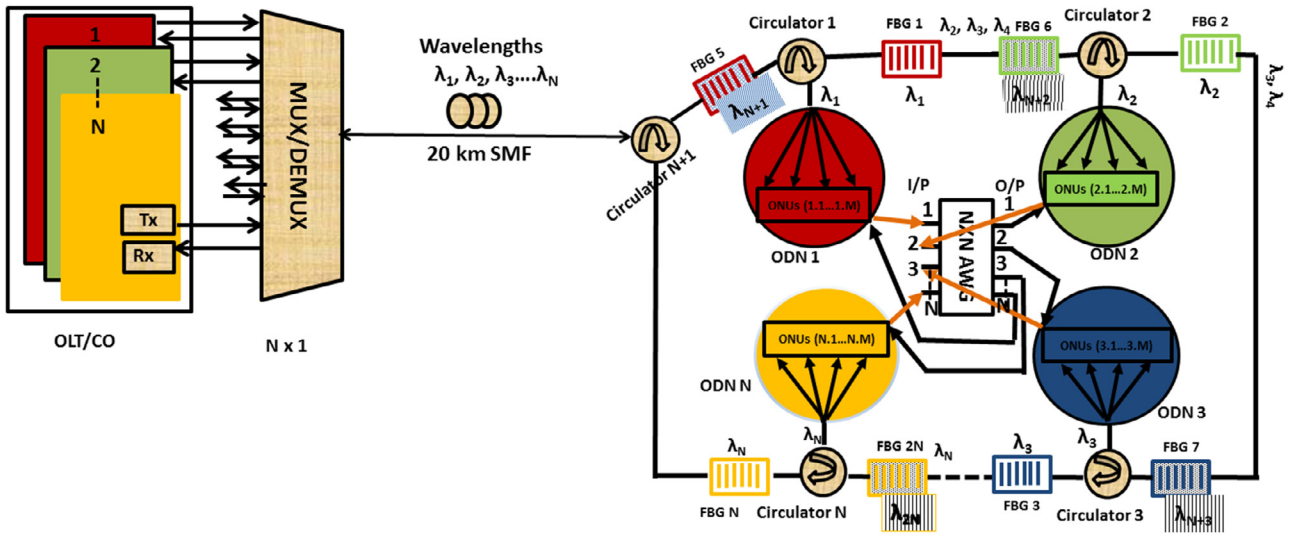


Fig. 1. Proposed Ring Based WDM-PON network.

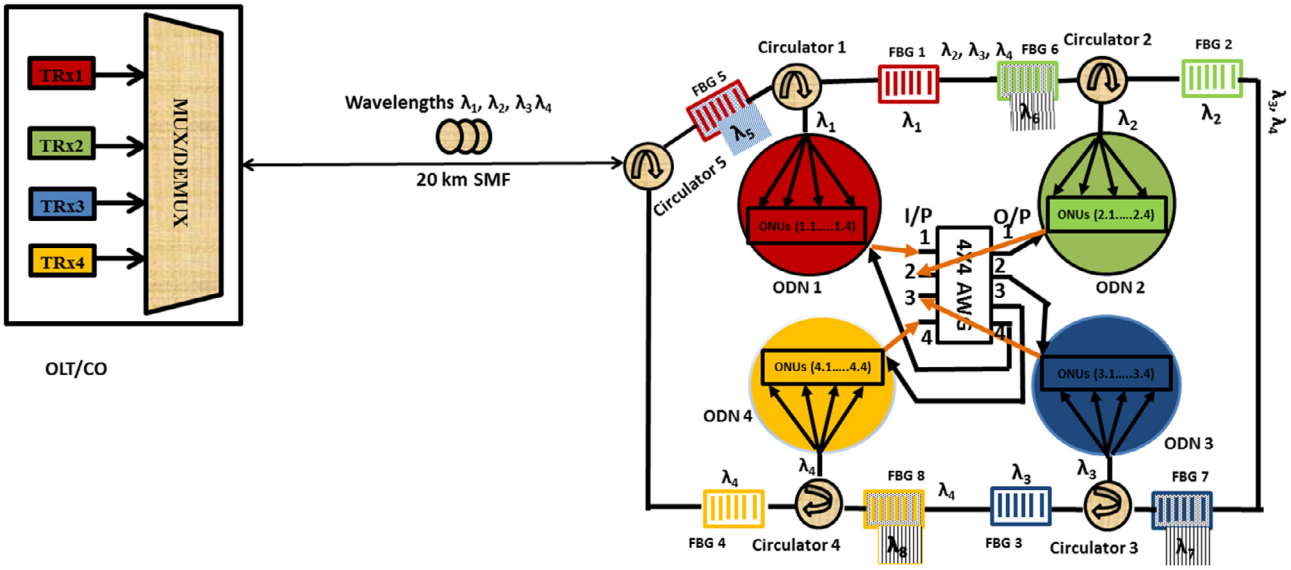


Fig. 2. Downstream transmission in a proposed ring architecture.

(both at user end and OLT side) and more bandwidth consumption (both upstream and downstream bandwidth) [4]. Therefore, this conventional approach may not be helpful for the development of smart cities as it consumes more power, more bandwidth and imposes more burden on the fiber.

Earlier, numerous direct inter-ONU communication approaches in WDM-PON have been reported. Some approaches reported broadcasting in which data from source ONU is transmitted to all ONUs while some approaches reported virtual private group (VPG) networking in which source ONU is sharing the data to some specific group of ONUs [5]. The VPG can be accomplished by employing subcarrier multiplexing [6], electronic code division multiple access [7], and reflected waveband grouping techniques [8] etc. Reflective semiconductor optical amplifier (RSOA) at ONU and fiber Bragg grating (FBG) at remote node (RN) may be employed to evade the external seeding of source and reflecting the wavelength from the RN itself towards the destined ONUs [9]. For dynamic routing of a wavelength to particular ONU, an arrayed waveguide grating (AWG) may also be incorporated at RN [10]. By transmitting the inter ONU data with tunable source and due to the cyclic property of the AWG, an output port of AWG is changed depending on its input wavelength. By tuning the transmitting ONU at proper wavelength, the destined ONU can be changed [11]. WDM-PON with the broadcast facility with local area services is proposed in [12], however, such architecture is not capable of providing inter-ONU and Intra-ONU broadcasting.

Inter-ONU communication in time-division multiplexing (TDM) PON is achieved by cascading an FBG with a splitter at RN [13]. However, such architecture only broadcasts the data to all the ONUs available in the network. For larger networks, such architectures may not be useful due to higher insertion losses and security issues.

In this manuscript, we have proposed a ring based latency aware and energy efficient Hybrid WDM-TDM PON network which is responsible for transmitting point to point downstream, upstream, direct inter-ODN, intra-ODN signal. Direct inter-ODN and intra-ODN communication techniques have the capability to transmit the signal directly without interfering the central office and avoiding remodulation, retransmission of inter-ODN and intra-ODN signal at OLT. Therefore, proposed architecture reduces energy consumption, transmission latency and OPEX significantly. The architecture also reduces the upstream and downstream bandwidth consumption by transmitting the

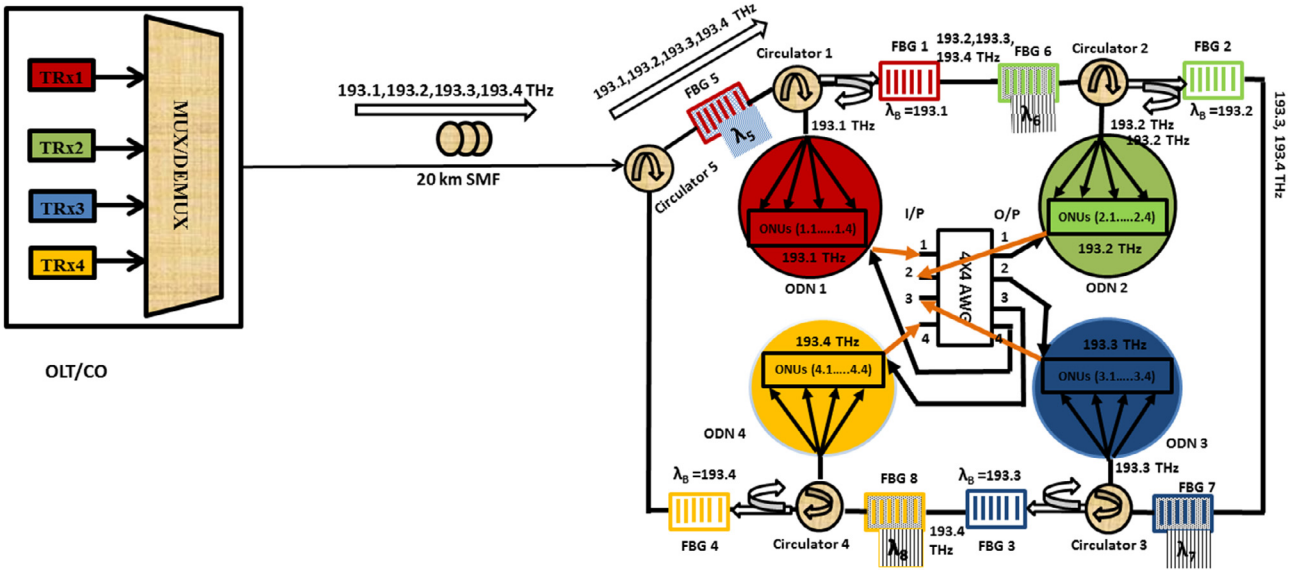


Fig. 3. Downstream transmission in the proposed architecture.

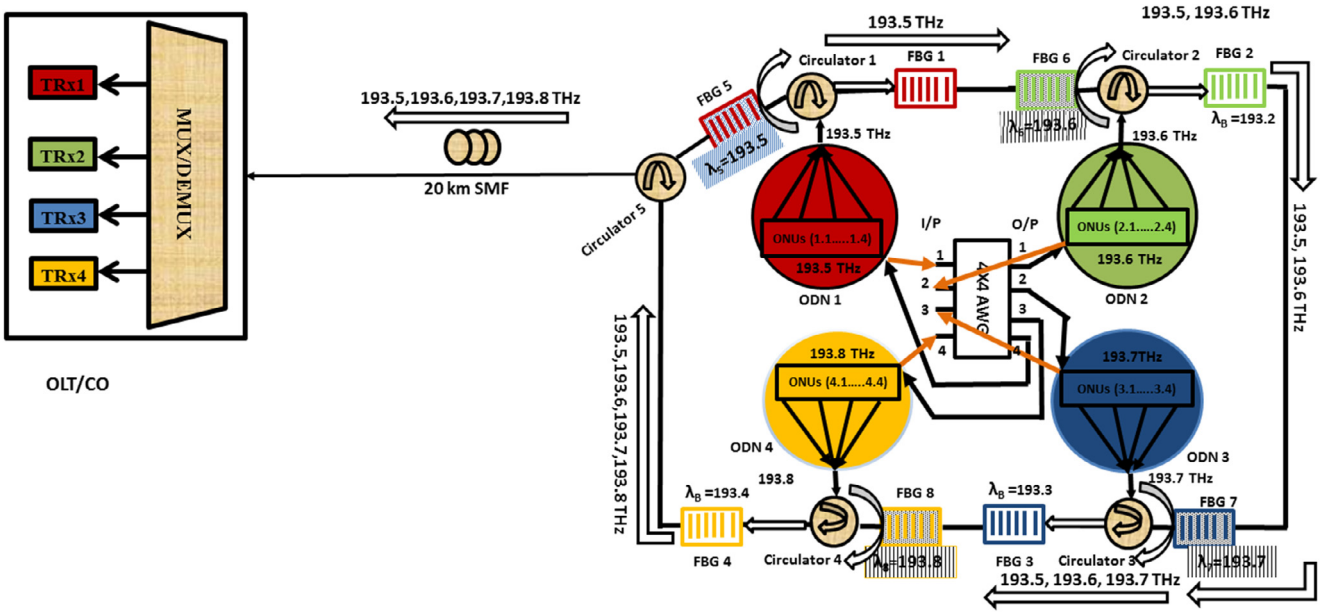


Fig. 4. Upstream transmission.

inter ODN signal locally. The proposed architecture also provides intra-ODN broadcasting and inter-ODN broadcasting which is its main unique feature and that further provides the opportunity to not send the same signal multiple times to multiple users located within the same group or other nearby located groups. To the best of our knowledge, all the above features are not available in any existing single architecture. The architecture will be useful for supporting and strengthening the ICT infrastructure for providing various high-speed applications of a smart city.

2. Proposed Architecture

The proposed ring based WDM-PON network is depicted in Fig. 1. At the central office (CO), the optical line terminal (OLT) has N transceivers for communication with N optical distribution networks (ODNs) connected in a circular ring at the user side. Furthermore, each ODN may serve up to M users connected in point to multipoint fashion, therefore the proposed network may support up to MN users. The signals are transmitted at different frequencies from the CW Laser. The transmitted power and the linewidth of the laser signal are kept at 0dBm (1mW) and 10 MHz respectively. The transmitted signals from CO are combined by Nx1 MUX and then transferred on 20 km single-mode fiber (SMF). The SMF has the attenuation losses of 0.2 dB/km and dispersion of 16.75 ps/nm/km. The dispersion slope of the fiber is considered as 0.075 ps/nm^2/km. These signals are then transferred to the circular ring as shown in Fig. 1. The ring part of the architecture consists of (N + 1) three-port circulators, 2 N fiber Bragg gratings (FBGs), N power splitters and one NxN AWG to support N ODNs connected in the ring. The main purpose of the three-port circulator is to transmit the signal from port1 to port2, port2 to port3 and port3 to port1. The insertion loss of the three-port circulator is considered as 0.5 dB. The FBG has a different Bragg wavelength depending upon the wavelength which is to be reflected. The insertion loss of FBG is approximately 0.5 dB. The Bandwidth of the FBG is 125 GHz for the 1550 nm window. The NxN AWG has a center frequency of 193.1 THz and bandwidth of 10 GHz. The frequency spacing of two consecutive

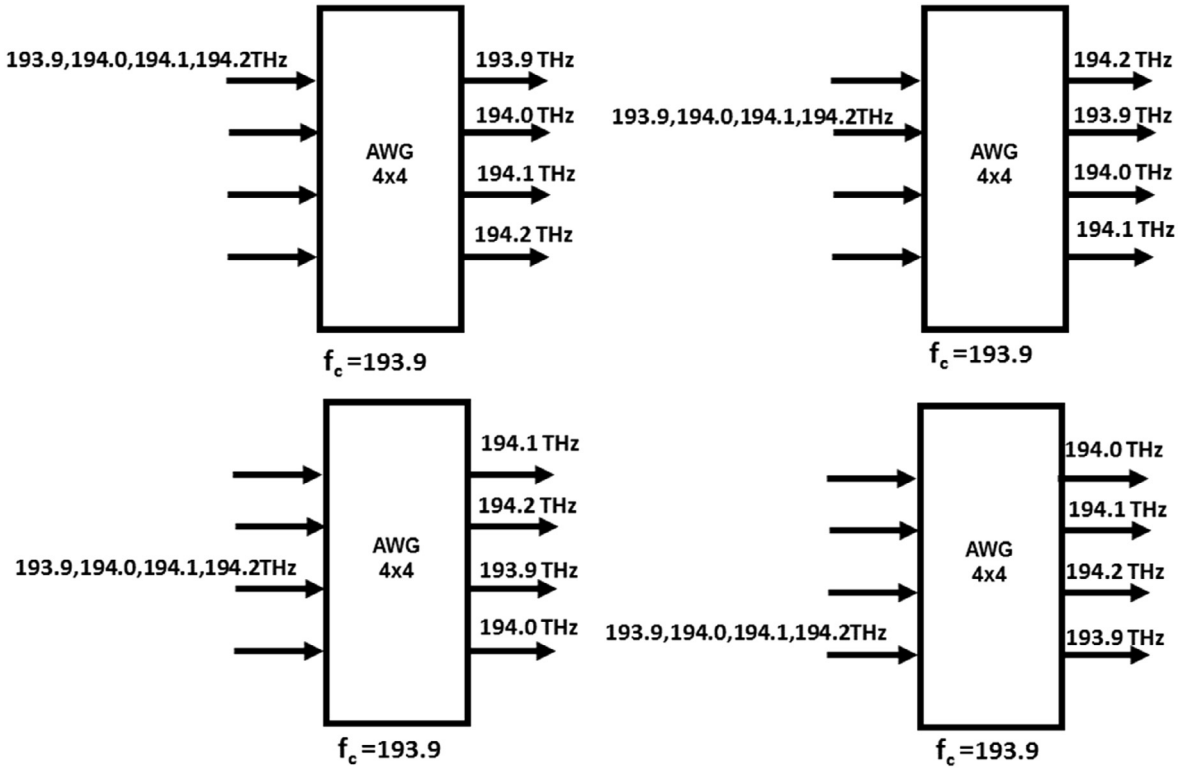


Fig. 5. Wavelength distribution for 4 × 4 AWG for input at different input ports.

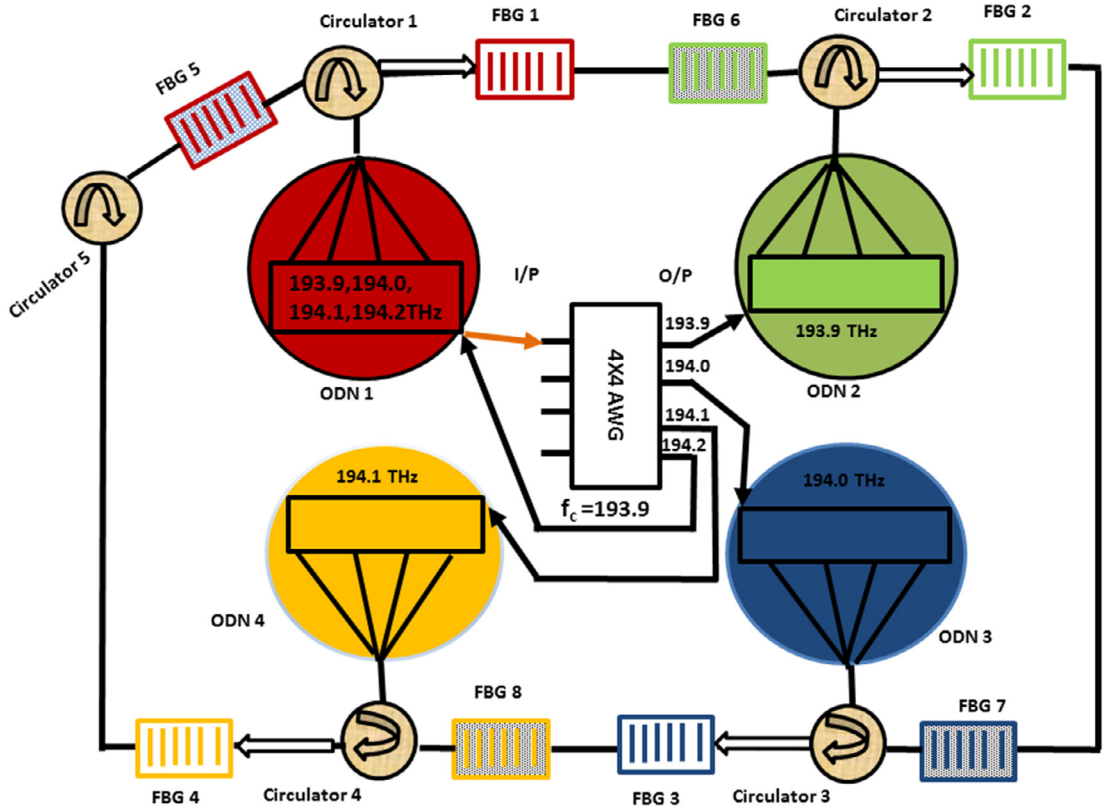


Fig. 6. ODN1 communicating with other ODNs.

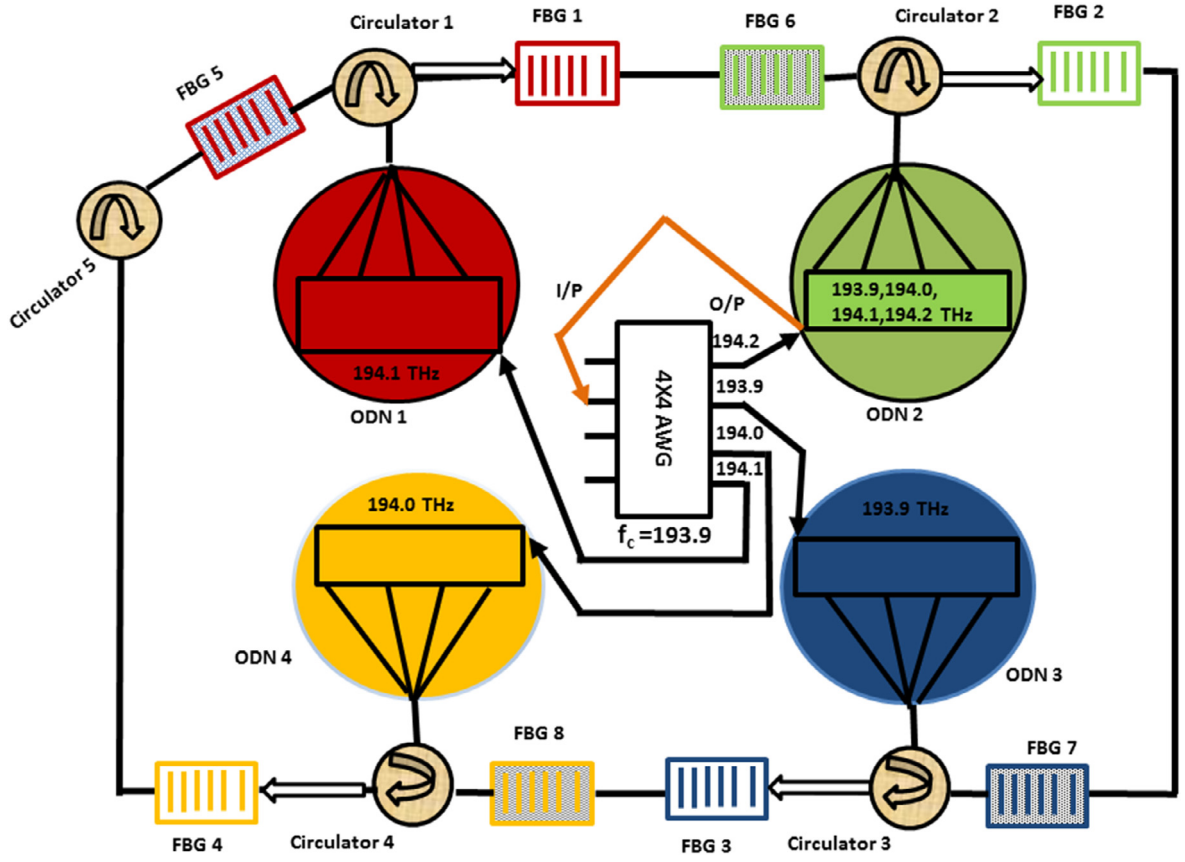


Fig. 7. ODN2 communicating with other ODNs.

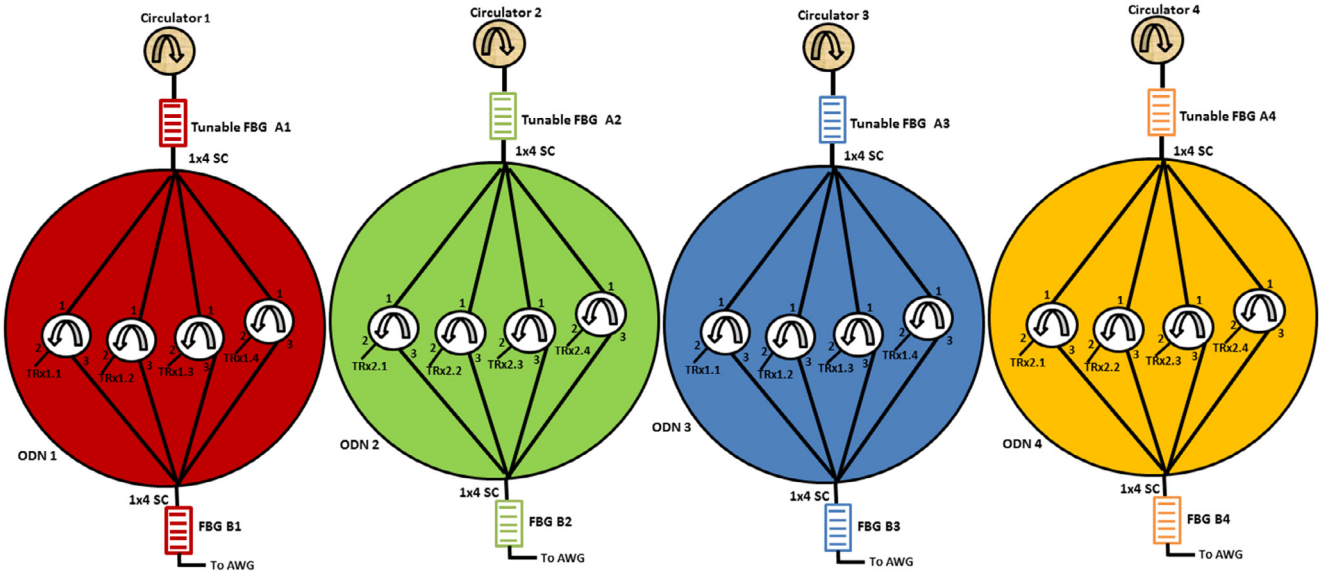


Fig. 8. Internal ODN structure.

ports of AWG is 100 GHz. The insertion loss of AWG is 5 dB. The insertion loss for power splitter of 1 × 4 size is approximately 7 dB. The detailed working of the architecture is explained in the next section.

3. Working and operation

The working of the proposed architecture is explained in Fig. 2. For the ease of explanation, we considered four ODNs and each ODN is further having four ONUs. i.e. the designed network is capable of handling upto sixteen ONUs at a time. OLT is also designed with four transceivers to transmit four downstream wavelengths λ_1, λ_2, λ_3 and λ_4 respectively for ODN1, ODN2, ODN3 and ODN4. At the Central office, these wavelengths are combined by a 4 × 1 AWG and transmitted to the ring through 20 km SMF. All ODNs are connected in a ring and the ring consists of five 3-port circulators, eight FBGs and four 1 × 4 power splitters as shown in Fig. 2.

Let downstream signals are transmitted at wavelengths 193.1 THz (λ_1), 193.2 THz (λ_2), 193.3 THz (λ_3) and 193.4 THz (λ_4) from

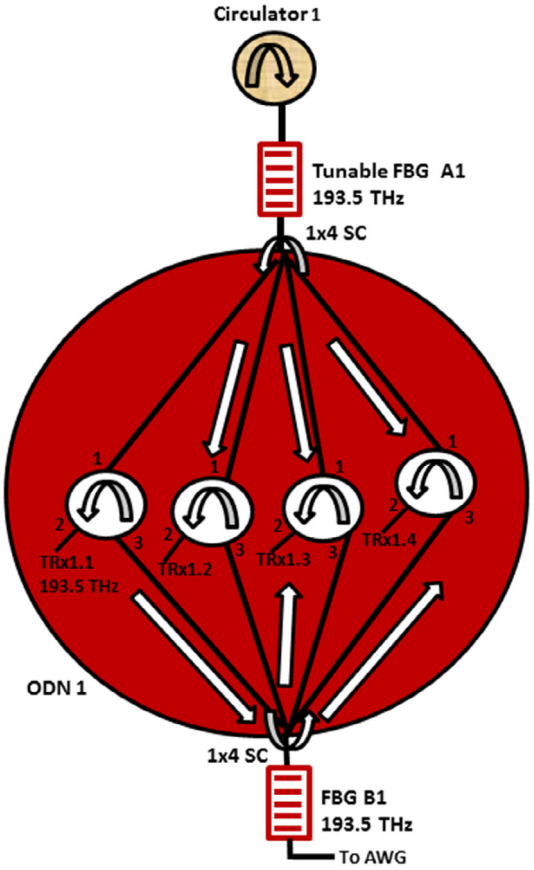


Fig. 9. Intra-ODN broadcasting.

transmitters TRx1, TRx2, TRx3, and TRx4 respectively. All these wavelengths are combined by the combiner and reach to port-1 of circulator-5 and then circulated to port-2 of the circulator. The Bragg wavelengths of FBG1, FBG2, FBG3 and FBG4 are kept at λ_1, λ_2, λ_3 and λ_4 i.e. 193.1 THz, 193.2 THz, 193.3 THz and 193.4 THz respectively for reflecting the respective downstream wavelength from FBG1, FBG2, FBG3 and FBG4 to provide services to ODN1, ODN2, ODN3 and ODN4 respectively. The Bragg wavelengths of FBG5, FBG6, FBG7 and FBG8 are set to λ_5, λ_6, λ_7 and λ_8 so these FBGs will reflect the upstream wavelengths λ_5, λ_6, λ_7 and λ_8 respectively and transmit the other wavelengths.

The stream of wavelengths 193.1 THz (λ_1), 193.2 THz (λ_2), 193.3 THz (λ_3) and 193.4 THz (λ_4) coming from port-2 of circulator-5 is passed through FBG5 and reaching to FBG1 via circulator-1. Due to the Bragg reflection, the wavelength 193.1 THz (λ_1) is reflected back by FBG1 and remaining wavelengths (193.2 THz (λ_2), 193.3 THz (λ_3) and 193.4 THz (λ_4)) are transmitted by FBG1 and reaching to FBG2 via FBG6 and circulator-2 as shown in Fig. 3. The reflected wavelengths 193.1 THz (λ_1) from FBG1 is now circulated to port-3 of circulator-1 and given to 1 × 4 power splitter for servicing the various ONUs located in ODN1 i.e. ONU1.1, ONU1.2, ONU1.3 and ONU1.4. Due to the Bragg reflection, the downstream wavelengths 193.2 THz (λ_2), 193.3 THz (λ_3) and 193.4 THz (λ_4) are also reflected by FBG2, FBG3 and FBG4 and servicing the ONUs of ODN2, ODN3 and ODN4 respectively.

The architecture also supports the upstream transmissions and let ODN1, ODN2, ODN3, and ODN4 transmit the signals at wavelengths 193.5 THz (λ_5), 193.6 THz (λ_6), 193.7 THz (λ_7) and 193.8 THz (λ_8) respectively in time division multiple access (TDMA) technique [14]. The ONUs located in the same ODN may also transmit the upstream data in the TDMA technique to avoid the simultaneous transmissions and collisions. The wavelengths 193.5 THz (λ_5) transmitted from ODN1 is circulated to port-1 of circulator-1 and reflected back from FBG5. Therefore, the reflected wavelength is now transmitted in the clockwise direction on the ring and finally reaching to the port-3 of the circulator-5 from where the upstream signal is diverted port-1 and then to the optical fiber and CO. Similarly, wavelengths 193.6 THz (λ_6), 193.7 THz (λ_7) and 193.8 THz (λ_8) are transmitted from ODN2, ODN3 and ODN4 for upstream transmission as shown in Fig. 4. Arrow marks in the figure indicate the upstream signal flow. At OLT, these wavelengths are separated by 1 × 4 De-mux and respective wavelengths are received by the receiver of TRx1, TRx2, TRx3 and TRx4 respectively.

Apart from the regular downstream and upstream transmissions, the proposed architecture is also capable of transmitting direct inter-ODN

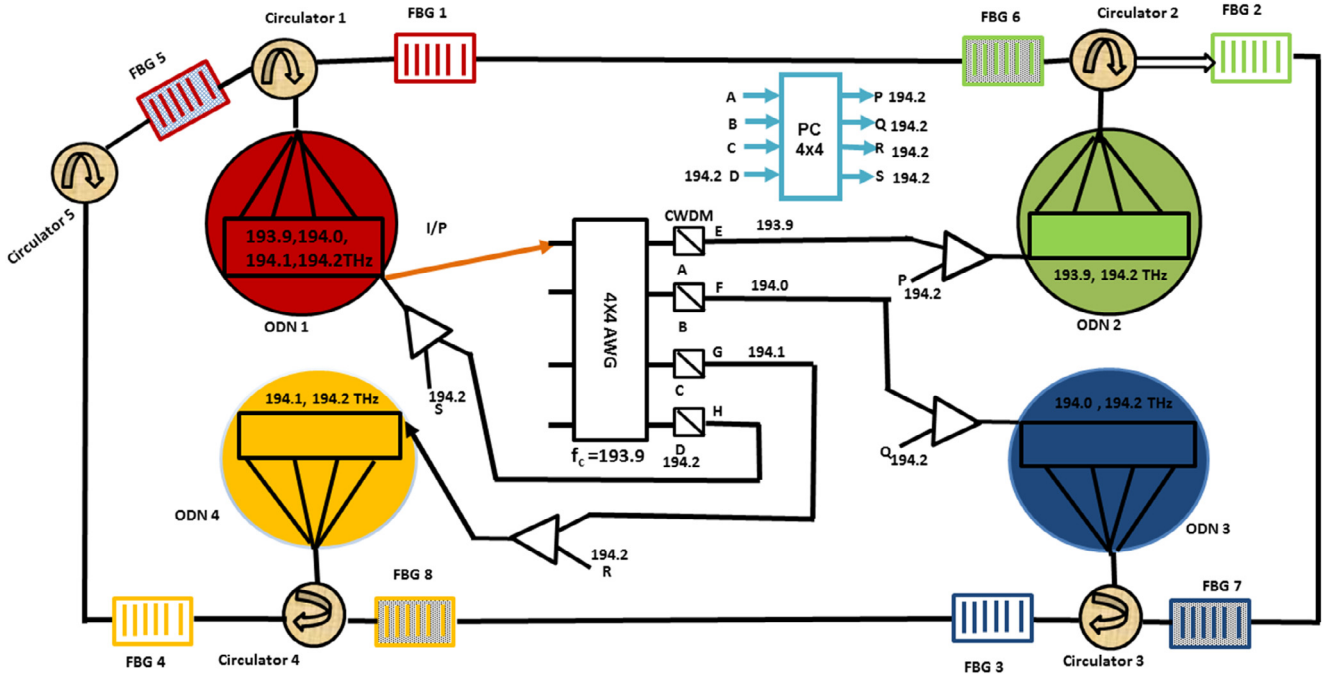


Fig. 10. ODN1 broadcasting to other ODNs.

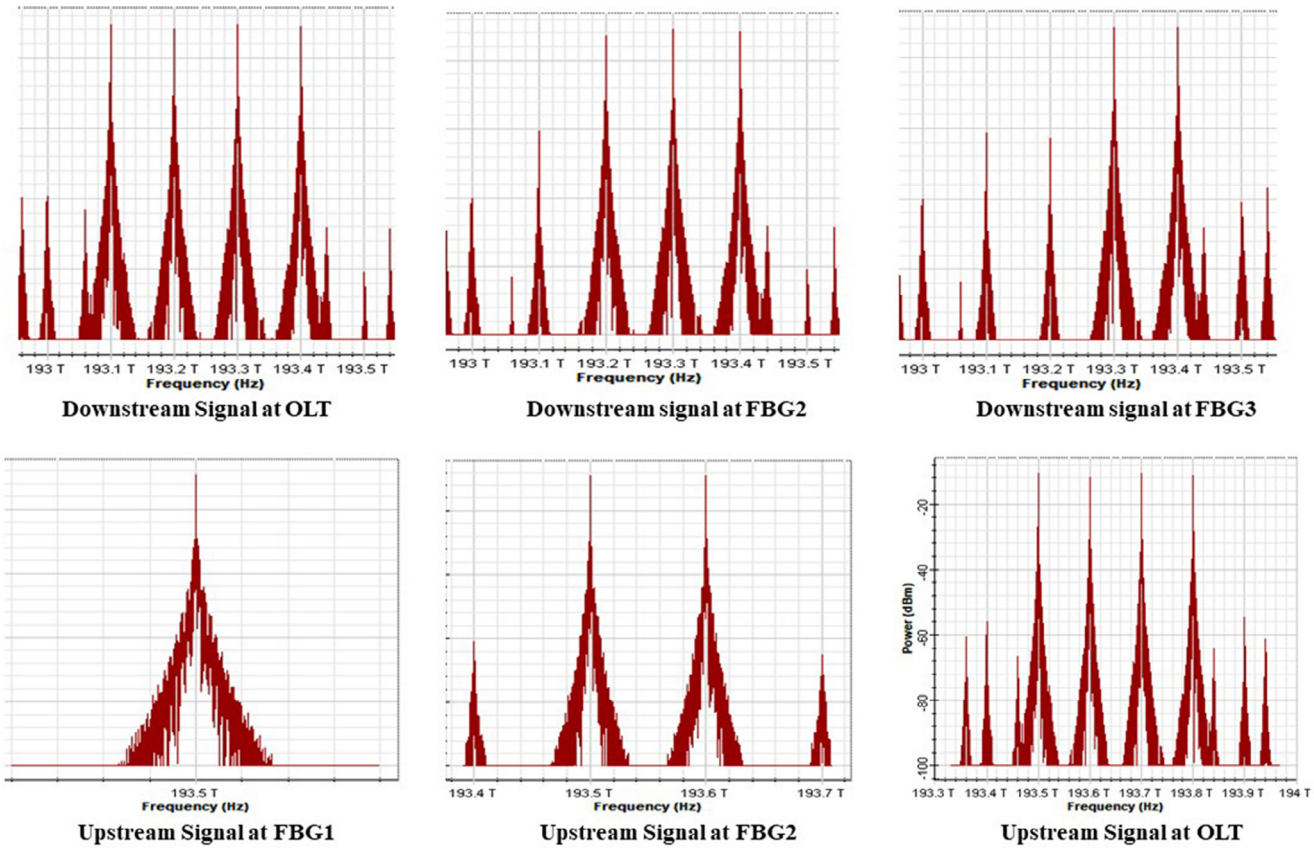


Fig. 11. Frequency domain representation of downstream and upstream signals at different points.

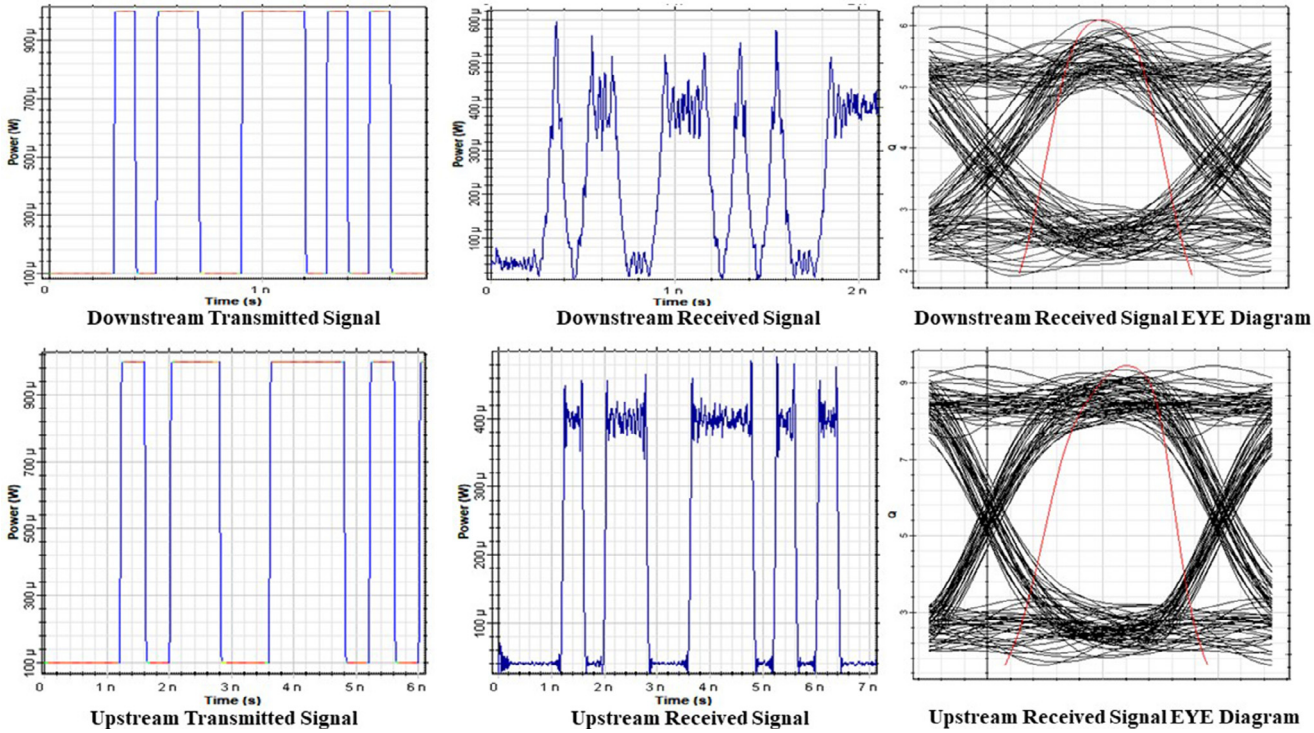


Fig. 12. Time-domain representation and received signal EYE diagram for Downstream and upstream signals.

communication signals and ODN broadcasting. For direct inter ODN communication, the tunable transmitter of all the ONUs located in the ODN1 are connected to the 4 × 1 combiner and then to input port1 of 4 × 4 AWG. Similarly, transmitters of ODN2, ODN3 and ODN4 are connected to input port2, port3 and port4 respectively. Output port1, port2, port3 and port4 are connected to ONUs (through 1 × 4 splitter) located in ODN2, ODN3, ODN4 and ODN1 respectively. If the inter-ODN signal is transmitted at any one or more wavelengths 193.9, 194.0,

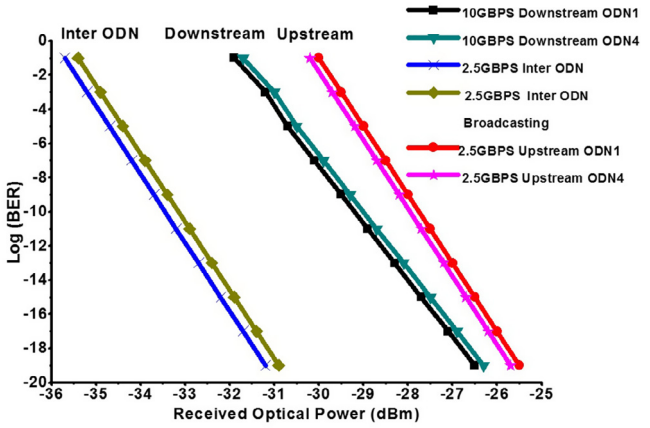


Fig. 13. BER performance for inter ODN, downstream and upstream transmission.

Table 1
Insertion loss for different network components.

Components	Insertion loss (dB)
SMF	0.2 dB/km
AWG	5 dB
Power splitter 1 × 4/4 × 4	7 dB
Power splitter/combiner 1 × 2	3.5 dB
CWDM	0.5 dB
Circulator	0.5 dB
FBG Transmission/reflection loss	< 0.5 dB

194.1 and 194.2 THz and if the center frequency of the AWG is kept 193.9 THz then due to the cyclic property of AWG [15], the transmitted signals at wavelengths 193.9, 194.0, 194.1 and 194.2 THz will reach to output port port1, port2, port3 and port4 respectively as shown in Fig. 5. If these wavelengths are applied to input port2 and center frequency of AWG remains the same as 193.9 THz then transmitted signals at wavelengths 193.9, 194.0, 194.1 and 194.2 THz will reach to output port port2, port3, port4 and port1 respectively. A total of four possible cases are depicted in Fig. 5. Therefore, in this manner, each ODN is capable of communicating directly with other ODN just by tuning the transmitted wavelength.

Direct Inter-ODN communication of ODN1 with other ODNs is depicted in Fig. 6. Let ODN1 want to communicate directly to the ODN2, ODN3 and ODN4. For this, at a time ODN1 transmits the signal at any one of the wavelengths from 193.9, 194.0, 194.1 and 194.2 THz depending on with which ODN, ODN1 wants to communicate directly. The center frequency of AWG is kept 193.9 THz, and due to the cyclic property of AWG, out of these four wavelengths, 193.9, 194.0, 194.1 and 194.2 THz will serve ODN2, ODN3, ODN4 and ODN1 respectively as clearly depicted in Fig. 6. Similarly, ODN2, ODN3 and ODN4 may also communicate directly with other ODNs by providing the inputs to port2, port3 and port4 respectively. Fig. 7 shows the inter ODN communication of ODN2 directly with other ODNs.

The internal structure of the ODNs is depicted in Fig. 8. Each ONU has a tunable transceiver (TRx) which is connected to the port-2 of the three-port circulator. These tunable TRx can be tuned at any particular wavelength to transmit/ receive the signal. Port-1 of all the circulators available inside an ODN is connected to the 1 × 4 star coupler, tunable FBG A and to the main ring. Port-3 of all the circulators is connected to another 1 × 4 star coupler, FBG B and to the AWG. For upstream transmission, the Bragg wavelength of the FBG B is kept similar to the upstream wavelength (to reflect the signal) and tunable FBG A is tuned at any other wavelength to transmit the upstream signal reflected from FBG B. For direct inter ODN communication, the Bragg wavelength of tunable FBG A is tuned in such a way that it reflects the inter ODN wavelength while FBG B1, FBG B2, FBG B3 and FBG B4 are fixed at wavelengths 193.5 THz (λ_5), 193.6 THz (λ_6), 193.7 THz (λ_7) and 193.8 THz (λ_8) respectively to transmit and provide the inter ODN wavelength to the input port of 4 × 4 AWG from where the signal is transmitted to the other ODNs. For example, Let ODN1 wants to transmit the data to ODN2 (ODN3, ODN4), for this ODN1 transmits the data at 193.9 THz (194.0 THz, 194.1 THz). The transmitted data is first circulated to port-3 and reached to FBG B1 which is having Bragg wavelength of 193.5 THz. As the center frequency of AWG is 193.9, the transmitted data 193.9 THz (194.0 THz, 194.1 THz) will reach to ODN2 (ODN3, ODN4). Inside the ODN, this inter ODN data is reached at port 3 of the circulator from where it is circulated to port 1 and to FGB. As FBG A2 (FBG A3 and FBG A4) is tuned at 193.9 THz (194.0 THz, 194.1 THz), the inter ODN signal reflected by respective FBG A and received by the transceivers.

The proposed architecture also provides intra-ODN communication i.e. the communication between the ONUs located inside an ODN. For this, both FBG A1 and FBG B1 are tuned at wavelength 193.5 THz, and the intra ODN signal is also transmitted at the same wavelength 193.5 THz. Let's say TRx1.1 transmits the data at 193.5 THz and which will be reflected from FBG B1 and reached to the FBG A1 via the circulator. Now, this wavelength is again reflected and reached to the respective transceivers of different ONUs as depicted in Fig. 9.

The proposed architecture is also capable of broadcasting the direct inter-ODN signal to all the ODNs available in the network. To achieve the inter-ODN communication and broadcasting, each output port of the AWG is cascaded with CWDM, 4 × 4 power coupler and 2 × 1 power combiner as shown in Fig. 10. The CWDM is used to separate out the 193.9, 194.0, 194.1 THz wavelengths from 194.2 THz. The CWDM transmit the wavelengths 193.9, 194.0, 194.1 THz to point E, F, G respectively and broadcasting wavelength 194.2 THz is diverted to port D. Point E, F, G and H are directly connected to one input port of each 2 × 1 power combiner while points A, B, C, D are connected to four input ports of 4 × 4 power coupler. Output ports of this coupler are connected to the another input port of each 2 × 1 power combiner. For the simplicity, the connections are not drawn in the figure rather indicated by points. In Fig. 10, the wavelengths 193.9, 194.0 and 194.1 THz are transmitting data directly to ODN2, ODN3 and ODN 4 respectively and wavelength 193.2 THz is diverted to port D which is splitted via 4 × 4 PC and applied to each ODN (through point P, Q, R, and S) for broadcasting purpose.

Table 2
A speculative case of a university with registrar office and four departments.

	Registrar Office (CO/OLT)	Electronics Engineering (ODN1)	Computer Engineering (ODN2)	Civil Engineering (ODN3)	Mechanical Engineering (ODN4)
Registrar Office (CO/OLT)		193.1 THz	193.2 THz	193.3 THz	193.4 THz
Electronics Engineering (ODN1)	193.5 THz		193.9 THz	194.0 THz	194.1 THz
Computer Engineering (ODN2)	193.6 THz	194.1 THz		193.9 THz	194.0 THz
Civil Engineering (ODN3)	193.7 THz	194.0 THz	194.1 THz		193.9 THz
Mechanical Engineering (ODN4)	193.8 THz	193.9 THz	194.0 THz	194.1 THz	

Table 3
Distance between circulator5 to different ODNs.

	Electronics Engineering (ODN1)	Computer Engineering (ODN2)	Civil Engineering (ODN3)	Mechanical Engineering (ODN4)
Circulator5 (Downstream Transmission)	100 m	300 m	500 m	700 m
Circulator5 (Upstream Transmission)	700 m	500 m	300 m	100 m

Table 4
Latency for various path components.

Components	Latency
20 km SMF	97.94 μs
100 m SMF	0.4897 μs
300 m SMF	1.469 μs
500 m SMF	2.448 μs
700 m SMF	3.427 μs
AWG	5 ns
Power splitter	1 ns
CWDM	5 ns
Circulator	5 ns
FBG Transmission/Reflection	< 5ns

4. Results & Discussions

The performance of the given network has been verified and confirmed with simulation results using OptiSystem 16.1. For the simulation purpose, we have considered four transmitters at OLT and four ODNs. Each ODN further consists of four ONUs. Frequency domain representation of Downstream and Upstream signals at different points including OLT and different FBGs are given in Fig. 11. At OLT all four wavelengths 193.1 THz, 193.2 THz, 193.3 THz and 193.4 THz are transmitted. As FBG2 is tuned at Bragg wavelength 193.1 THz, it reflects the wavelength 193.1 THz and the remaining three wavelengths are received. Similarly, FBG3 reflects the 193.2 THz wavelength and two wavelengths are received and so on.

Time-domain representation of transmitted and received signal for downstream and upstream direction is given in Fig. 12. EYE diagram of the received signal for both Downstream and Upstream directions are also depicted in Fig. 12, which indicates the clear reception of the transmitted signals.

BER performance of the proposed architecture for different cases including back-to-back (B2B), downstream, upstream and inter-ODN transmission at different data rates are given in Fig. 13. The performance for different received optical powers clearly indicates that the direct inter-ODN signal needs roughly 6 dB and 4 dB less optical powers at receiver as compared to traditional downstream and upstream transmissions respectively. The power penalty occurring in the traditional transmissions is due to the various nonlinear effects occurring in the system. At the same time, for direct inter-ODN transmission, the signal has to travel only a few meters rather than the kilometers and therefore nonlinear effects are insignificant in the proposed mechanism. For traditional downstream and upstream transmissions, cumulative fiber losses are more and to maintain the minimum receiver sensitivity, a significant amount of higher power needs to be transmitted. Also, for the higher input powers above a certain threshold, fiber nonlinear effects become significant and therefore degrades the system's performance. The input power required for direct inter-ODN communication is very less and therefore provides the opportunity to save significant operation power with a better quality of transmission. Numerical values of insertion losses corresponding to various network components are given in Table 1.

To explain the proposed architecture clearly, we have considered a speculative example of a university in which there is a registrar office and four departments and they may communicate with each other. Let registrar office is located at CO/OLT and four departments namely Electronics Engineering, Computer Engineering, Civil Engineering, and Mechanical Engineering are located at ODN1, ODN2, ODN3 and ODN4 respectively. To communicate the information from the registrar office to various departments or vice versa or from one department to another department directly, various possible wavelengths are given in Table 2. For example, a message from the registrar office to various departments namely Electronics Engineering, Computer Engineering, Civil Engineering and Mechanical Engineering can be sent in downstream direction at wavelength 193.1 THz, 193.2 THz, 193.3 THz and 193.4 THz respectively. For upstream direction, Electronics Engineering, Computer Engineering, Civil Engineering and Mechanical Engineering departments can send information to the registrar office on 193.5 THz, 193.6 THz, 193.7 THz, 193.8 THz respectively. For direct inter-ODN communication, any department can send the data to other departments on a particular wavelength given in Table 2. For example, the Electronics Engineering department can send the information to Computer Engineering, Civil Engineering and Mechanical Engineering departments at 193.9 THz, 194.0 THz and 194.1 THz respectively.

Let the Circulator1, Circulator2, Circulator3 and Circulator4 connected to the main ring are placed at 100 m distance consecutively. i.e. Electronics Engineering, Computer Engineering, Civil Engineering and Mechanical Engineering departments placed at a distance of 100 m consecutively. Also, let these departments are connected to the 4 × 4 AWG and the fiber distances between from each department to AWG is 100 m. Downstream and upstream transmissions are carried out on the main circular ring and inter ODN communication is carried out through the AWG. Path distance for upstream and downstream transmissions from circulator5 (from where the ring is connected to the mainline and OLT) to destined/source ODN are depicted in Table 3.

For analyzing the latencies for downstream, upstream and inter ODN transmission, we need to consider the latencies occurring due to fiber (path) and various components arriving in that path. Fiber latency can be calculated by the effective velocity of the optical signal in the fiber which can be given by Eq. (1). Where C is the velocity of light in vacuum and n is the effective refractive index of the fiber. At wavelength 1550 nm, the effective refractive index is 1.4682 [16] and thus, effective fiber velocity is-

Table 5
Latency calculations for different transmissions.

	Registrar office (CO/OLT)	Electronics Engineering (ODN1)	Computer engineering (ODN2)	Civil engineering (ODN3)	Mechanical engineering (ODN4)
Registrar office (CO/OLT)		98.45 μs	98.94 μs	99.45 μs	99.95 μs
Electronics engineering (ODN1)	99.95 μs		1.006 μs	1.006 μs	1.006 μs
Computer engineering (ODN2)	99.45 μs	1.006 μs		1.006 μs	1.006 μs
Civil engineering (ODN3)	98.94 μs	1.006 μs	1.006 μs		1.006 μs
Mechanical engineering (ODN4)	98.45 μs	1.006 μs	1.006 μs	1.006 μs	

Table 6
Engagement duration (hours) between various offices.

	Registrar office (CO/OLT)	Electronics engineering (ODN1)	Computer engineering (ODN2)	Civil engineering (ODN3)	Mechanical engineering (ODN4)
Registrar office (CO/OLT)		5	6	6	7
Electronics engineering (ODN1)	5		6	6	7
Computer engineering (ODN2)	6	6		7	5
Civil engineering (ODN3)	6	6	7		5
Mechanical engineering (ODN4)	7	7	5	5	

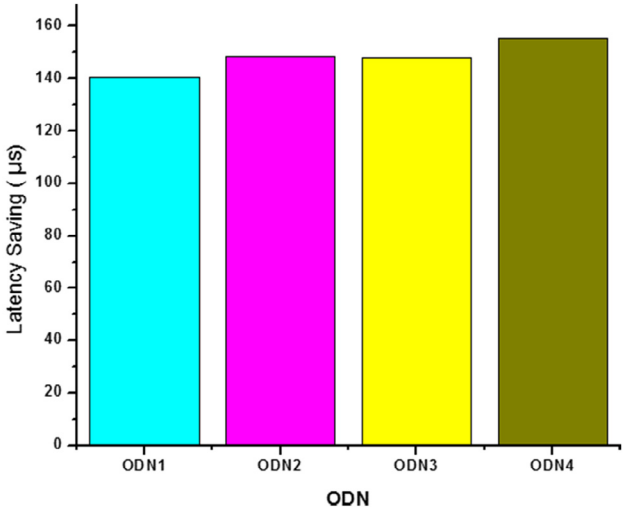


Fig.14. Latency saving for various ODNs.

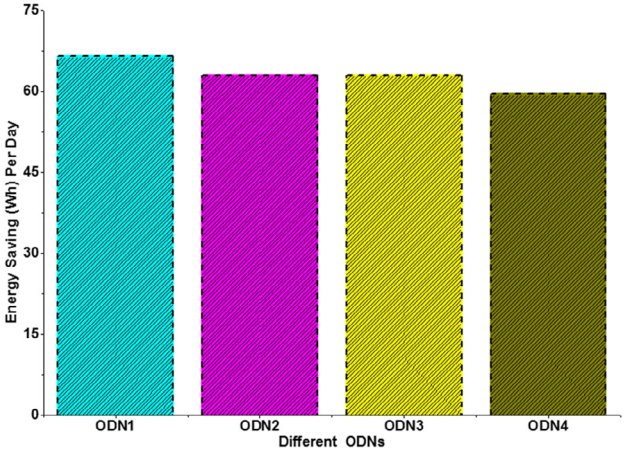


Fig. 15. Energy-saving (Wh) per hour for various ODNs.

$$v = \frac{c}{n} = \frac{299792.45km/s}{1.4682} = 204190.477km/s \quad (1)$$

Now, the fiber transmission latency for different transmission lengths can be calculated by dividing the fiber length by effective velocity. The latencies for various used components and fiber lengths are given in Table 4. Fiber latencies are much higher than other components and the minimization of fiber latencies gives the reduces the effective transmission latency significantly.

Latencies occurring for the transmission of the downstream data from OLT to various ODNs may be calculated by adding the latencies of individual components occurring the path and may be given by Eqs. [2–5] below: -

$$t_{OLT-ODN1} = t_{AWG(CO)} + t_{Fiber(20km)} + t_{Cir(N+1)} + t_{Fiber(100m)} + 2t_{Cir(1)} + 2 t_{Fiber(Cir1-FBG1)} \quad (2)$$

$$t_{OLT-ODN2} = t_{AWG(CO)} + t_{Fiber(20km)} + t_{Cir(N+1)} + t_{Fiber(100m)} + t_{Cir(1)} + t_{Fiber(Cir1-Cir2)} + 2t_{Cir(2)} + 2t_{Fiber(Cir2-FBG2)} \quad (3)$$

$$t_{OLT-ODN3} = t_{AWG(CO)} + t_{Fiber(20km)} + t_{Cir(N+1)} + t_{Fiber(100m)} + t_{Cir(1)} + t_{Fiber(Cir1-Cir2)} + t_{Cir(2)} + t_{Fiber(Cir2-Cir3)} + 2t_{Cir(3)} + 2t_{Fiber(Cir3-FBG3)} \quad (4)$$

$$t_{OLT-ODN4} = t_{AWG(CO)} + t_{Fiber(20km)} + t_{Cir(N+1)} + t_{Fiber(100m)} + t_{Cir(1)} + t_{Fiber(Cir1-Cir2)} + t_{Cir(2)} + t_{Fiber(Cir2-Cir3)} + t_{Cir(3)} + t_{Fiber(Cir3-Cir4)} + 2t_{Cir(4)} + 2 t_{Fiber(Cir4-FBG4)} \quad (5)$$

Similarly, latencies occurring for the transmission of the upstream data from various ODNs to OLT may be calculated by adding the latencies of individual components occurring in the path and may be given by Eqs. [6–9] below: -

$$t_{ODN1-OLT} = 2t_{Cir(1)} + 2t_{Fiber(Cir1-FBG(N+1))} + t_{Fiber(Cir1-Cir2)} + t_{Cir(2)} + t_{Fiber(Cir2-Cir3)} + t_{Cir(3)} + t_{Fiber(Cir3-Cir4)} + t_{Cir(4)} + t_{Fiber(Cir4-Cir(N+1))} + t_{Fiber(20km)} + t_{AWG(CO)} \quad (6)$$

$$t_{ODN2-OLT} = 2t_{Cir(2)} + 2t_{Fiber(Cir2-FBG(N+2))} + t_{Fiber(Cir1-Cir2)} + t_{Fiber(Cir2-Cir3)} + t_{Cir(3)} + t_{Fiber(Cir3-Cir4)} + t_{Cir(4)} + t_{Fiber(Cir4-Cir(N+1))} + t_{Fiber(20km)} + t_{AWG(CO)} \quad (7)$$

$$t_{ODN3-OLT} = 2t_{Cir(3)} + 2t_{Fiber(Cir3-FBG(N+3))} + t_{Fiber(Cir3-Cir4)} + t_{Cir(4)} + t_{Fiber(Cir4-Cir(N+1))} + t_{Fiber(20km)} + t_{AWG(CO)} \quad (8)$$

$$t_{ODN4-OLT} = 2t_{Cir(4)} + 2t_{Fiber(Cir4-FBG(2N))} + t_{Fiber(Cir4-Cir(N+1))} + t_{Fiber(20km)} + t_{AWG(CO)} \quad (9)$$

In the above expressions, $t_{OLT\text{-}ODN1}$ is the latency of downstream signal from OLT to ODN1. t_{AWG} is the AWG latency located at CO and $t_{fiber(20km)}$ is the latency of the 20 km fiber. $t_{ODN1\text{-}OLT}$ is the latency of upstream signal from ODN1 to OLT. t_{cir1} is the latency of the three-port circulator-1 and $t_{fiber(cir1\text{-}cir2)}$ is the latency between circulator1 to circulator2. $t_{fiber(cir1\text{-}FBG\ N+1)}$ is the fiber latency between circulator1 and FBG (N + 1). Similarly, other latencies may be assumed as per the subscripts.

The latencies for direct inter ODN communication from one ODN to another ODN and for broadcasting may be calculated by Eqs. (10), (11)

Table 7
Important benefits and limitations of similar hybrid PON architectures.

Paper	Energy saving	Latency awareness	Broadcasting from OLT	Inter ONU Broadcasting	Intra-ODN(subgroup of ONUs) Broadcasting	Limitations
Garg et al. [4]	High	Yes	Yes	Yes	No	No subgroup broadcasting
Q. Zhao et al. [5]	High	Yes	No	Yes	No	No subgroup broadcasting
N. Nadarajah et al. [6]	Low	Yes	Yes	Yes	No	No subgroup broadcasting, More RN losses
Nadarajah et al. [7]	High	Yes	Yes	Yes	No	No subgroup broadcasting
Chae et al. [9]	Low	Yes	Yes	Yes	No	No subgroup broadcasting, More RN losses
Garg et al. [10]	Moderate	Yes	Yes	Yes	Yes	Subgroup broadcasting, possible but no dedicated ONU inter ONU communication
Zhow et al. [12]	Moderate	Yes, Different latency for different ONUs connected in ring	No	No	No	More latency due to ring architecture
Yan et al. [13]	Yes	Yes	Yes	Yes	No	No subgroup broadcasting
Proposed	High	Yes, Same latency for all ONUs connected in ring for inter ODN communication	No	Yes	Yes	Subgroup broadcasting, possible but OLT Broadcasting is not possible

given below: -

$$t_{ODN(X)-ODN(Y)} = t_{CIR} + t_{SC} + t_{FBG} + t_{Fiber(100m)} + t_{4\times4AWG} + t_{Fiber(100m)} + t_{FBG} + t_{SC} + t_{CIR} \text{ Or}$$
$$t_{ODN(X)-ODN(Y)} = 2t_{CIR} + 2t_{SC} + 2t_{FBG} + 2t_{Fiber(100m)} + t_{4\times4AWG} \quad (10)$$

$$t_{Broadcasting\,[ODN(X)-ODN(Y)]} = 2t_{CIR} + 2t_{SC} + 2t_{FBG} + 2t_{Fiber(100m)} + t_{4\times4AWG} + t_{CWDM} + 2t_{SC/PC} \quad (11)$$

where X and Y represent the particular ODN and have the values ranging from 1 to N. Total path latency occurring between OLT to different ODNs, different ODNs to OLT and from one ODN to another ODN is given in Table 5. These latencies are given for one-time transmission between two points and will be scaled if the number of transmission increases. For inter ODN broadcasting, additional latency of approx. 7 ns will be present due to extra components CWDM and SC/PC.

Table 6 shows the representative number of transmission terms of one ODN with other ODNs/OLT in a day. For example, in a day, ODN1 communicates with OLT, ODN2, ODN3, ODN4 for 5, 6, 6, 7 times respectively. Also let each transmission is for 1 hour i.e. in a day, ODN1 communicates with OLT, ODN2, ODN3, ODN4 for 5, 6, 6, 7 hours respectively.

Latency saving for a specific ODNx can be calculated by subtracting the latencies occurring in the proposed direct inter-ODN approach from the conventional approach and can be given by-

$$t_{Saving}(ODNx) = \sum_{y=1\&y\neq x}^{4} (t_{ODNx-OLT-ODNy} - t_{ODNx-ODNy}) \quad (12)$$

$$t_{Saving}(ODNx) = 3(t_{ODNx-OLT}) + \sum_{y=1\&y\neq x}^{4} (t_{OLT-ODNy} - t_{ODNx-ODNy}) \quad (13)$$

where x represents the specific ODN for which latency efficiency is being calculated. y represents the destined ODN. $t_{ODNx\text{-}ODNy}$ represents the latency occurring for direct inter-ODN transmission from transmitting ODNx to destined ODNy. $t_{ODNx\text{-}OLT\text{-}ODNy}$ represents the total latency occurring for the conventional approach. From Equation (13) and various transmission data's as mentioned in Tables 5 and 6, the normalized latency savings per ODN per transmission is given in Fig. 14.

In the case of the conventional technique, the data is transmitted from the transmitted ODN to the OLT and then at OLT the data is re-modulated and retransmitted from the OLT to the destined ODN. While by implementing the proposed direct inter-ODN communication technique, the data needs to be transmitted only from the transmitting ODN and is able to reach the destined ODN without the use of any active components. Therefore, the proposed approach is capable of saving a significant amount of energy. Energy-saving for a specific ODNx can easily be calculated by subtracting the energy consumption by the proposed approach from the energy consumption of a conventional approach and can be given by equation (16).

$$E_{Saving}(ODNx) = \sum_{y=1\&y\neq x}^{4} (E_{ODNx-OLT-ODNy} - E_{ODNx-ODNy}) \quad (14)$$

$$E_{Saving}(ODNx) = 3\cdot E_{ODNx-OLT} + \sum_{y=1\&y\neq x}^{4} (E_{OLT-ODNy} - E_{ODNx-ODNy}) \quad (15)$$

$$E_{Saving}(ODNx) = 3\cdot P_{ODNx-OLT}\cdot h_{ODNx-OLT} + \sum_{y=1\&y\neq x}^{4} (P_{OLT-ODNy}\cdot h_{OLT-ODNy} - P_{ODNx-ODNy}\cdot h_{ODNx-ODNy}) \quad (16)$$

$E_{ODNx\text{-}OLT\text{-}ODNy}$ represents the total energy consumed for the transmission of a signal from ODNx to ODNy via OLT through the conventional approach. $E_{ODNx\text{-}ODNy}$ indicates the energy consumed to transmit the signal by using the proposed approach. $E_{ODNx\text{-}OLT}$ is the energy consumption for the transmission of a signal from ODNx to OLT. $E_{OLT\text{-}ODNy}$ is the energy consumption for the transmission of a signal from OLT to ODNy. P and h represent the power and time (hour) quantities for different scenarios as discussed for energy E.

For the conventional inter-ODN signal transmission, the signal is first transmitted from the source ODNx to OLT and then to the destined ODNy i.e. OLT is engaged most of the time and OLT line cards need to power ON which require approximately 3.5 W power. For the measured BER performance in Fig. 13, the minimum required receiver power is approximately −28 dBm (0.00158 mW) and −30 dBm (0.001 mW) for upstream and downstream transmission for minimum BER of 1×10^{-9}. While for direct inter-ODN transmission the minimum required receiver power is −34 dBm (0.0004 mW). From Eq. (16), the energy-saving per hour for the proposed inter-ODN communication for different ODNs is given in Fig. 15 below.

The proposed inter ODN transmission is capable of saving of average energy of up to 63 Wh per day per ODN for the assumed ODN engagements. For an ODNs, approximately 23 kWh of energy is saved for an entire year. As the number of users is increasing exponentially with the huge growth in ICT, for N ODNs, 23*N kWh power will be saved by the proposed architecture and will be responsible for reducing the significant emission of environmental hazardous gasses. Our proposed architecture has some advantages and some limitations as compared to the previously proposed architectures. The important benefits and limitations are compared in the Table 7.

Conclusion

A ring-based Hybrid WDM-TDM PON has been proposed which is capable of providing multiple transmissions such as the downstream, upstream, direct inter-ODN, intra-ODN, inter-ODN broadcasting and intra-ODN broadcasting. The performance of the proposed architecture for various transmissions is checked with the simulation tool and verified with the corresponding EYE diagrams, received power vs BER plot. The received power at the receiver is greater than the minimum required receiver power, i.e. approximately −28 dBm, −30 dBm and −34 dBm for upstream, downstream and direct inter-ODN transmission respectively for minimum BER of 1×10^{-9}. The proposed inter ODN transmission as compared to conventional transmission is saving average energy of up to 63 Wh per day per ODN approximately for the assumed ODN engagements. For N ODNs, approximately 23*N kWh of energy is saved for an entire year that is a huge amount for next-generation networks. The inter-ODN transmission is also reducing approximately 150 μs of latency per transmission for an ODN that is massive at 10GBPS data rates. The proposed architecture is responsible for strengthening the ICT infrastructure and reducing the significant emission of environmental hazardous gasses for developing smart cities.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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